Step 2: For each percentage p_n that is equal to or greater than $2p\%$, the percentage of time to be used in determining the propagation mode (1) path loss is p_v.

$$p_v = 100\, p / p_n \qquad \% \qquad \text{for } p_n \geq 2p\% \tag{125}$$

For each percentage of time, determine the distance, d_n (km), for which the propagation mode (1) predicted path loss is equal to the propagation mode (1) minimum required loss, using the propagation model in accordance with § 4 of the main body of this Appendix and the equation:

$$L_{bn}(p_v) = P_t + G_{en} + G_x - P_r(p) \qquad \text{dB} \tag{126}$$

The values of p_v must be within the range of percentage of time of the propagation mode (1) model (see § 1.5.1 of the main body of this Appendix).

Step 3: The propagation mode (1) required distance for the azimuth under consideration is the largest of the distances, d_n (km), calculated in Step 2, except when this largest distance is attained for the smallest value of p_n that is equal to or greater than $2p$ in accordance with equation (125) in Annex 6. In such cases, the propagation mode (1) required distance for the azimuth under consideration is the distance determined from equation (126) in Annex 6 with $G_{en} = G_{max}$ and $p_v = 50\%$ where G_{max} is the maximum value of horizon antenna gain.

Step 4: The propagation mode (1) supplementary contour distance for the azimuth under consideration is the required distance as determined in Step 3, except that the distance must be between the minimum coordination distance, d_{min}, and the maximum coordination distance, d_{max1}. These limits are given in § 4.2 and § 4.3 of the main body of this Appendix, respectively.

ANNEX 7

System parameters and predetermined coordination distances for determination of the coordination area around an earth station

1 Introduction

Tables 7 to 9 contain the system parameter values required by the methods in the main body of this Appendix to determine the coordination area around an earth station when the band is shared with terrestrial radiocommunication services or other earth stations operating in the opposite direction of transmission.

Table 7 is limited to those system parameter values required for the case of a transmitting earth station sharing with terrestrial services; Table 8 is limited to those parameter values required for the case of a receiving earth station sharing with terrestrial services; Table 9 is limited to those parameter values required for the case of a transmitting earth station which is sharing in a bidirectionally allocated band with other earth stations operating in the opposite direction of transmission.

These system parameter tables include primary allocations to the space and terrestrial services in Article **5** in all bands between 100 MHz and 105 GHz. Some of the columns have incomplete information. In some cases, this is because there is no requirement to calculate coordination distances as pre-determined coordination distances apply. In other cases, the service allocations are new and the systems may not be introduced for some years. Hence, the system parameters are the subject of ongoing development within the Radiocommunication Study Groups.

Parameters specific to the earth station, for which coordination is being sought, are provided to the Radiocommunication Bureau in the format specified in Appendix **4** as part of the notification and coordination procedures.

The row in each table entitled "method to be used" directs the user to the appropriate section of the main body of this Appendix which describes the methods to be followed for the determination of the coordination area.

Note that the earth station for which the coordination area is to be determined is identified by the service designation given in the first row of each table.

When a supplementary contour is to be developed, for example for digital fixed systems, the necessary system parameters may be found in one of the adjacent columns in Tables 7, 8 and 9. If no suitable system parameters are available, then the value of the permissible interference power ($P_r(p)$) may be calculated using equation (127) in § 2.

The predetermined coordination distances specified in Table 10 are used for transmitting and receiving earth stations, in cases defined by the corresponding frequency sharing situation.

2 Calculation of the permissible interference power of an interfering emission

Tables 7, 8 and 9 contain values for the parameters which are required for the calculation of the permissible interference power of the interfering emission (dBW), in the reference bandwidth, to be exceeded for no more than p% of the time at the receiving antenna terminal of a station subject to interference, from a single source of interference, using the general formula:

$$P_r(p) = 10 \log (k\, T_e\, B) + N_L + 10 \log (10^{M_s/10} - 1) - W \quad \text{dBW} \qquad (127)$$

where:

k: Boltzmann's constant (1.38×10^{-23} J/K)

T_e: thermal noise temperature of the receiving system (K), at the terminal of the receiving antenna (see § 2.1 of this Annex)

N_L: link noise contribution (see § 2.2 of this Annex)

B: reference bandwidth (Hz), i.e. the bandwidth in the receiving station that is subject to the interference and over which the power of the interfering emission can be averaged

p: percentage of the time during which the interference from one source may exceed the permissible interference power value; since the entries of interference are not likely to occur simultaneously, $p = p_0/n$

p_0: percentage of the time during which the interference from all sources may exceed the threshold value

n: number of equivalent, equal level, equal probability entries of interference, assumed to be uncorrelated for small percentages of the time

M_s: link performance margin (dB) (see § 2.3 of this Annex)

W: a thermal noise equivalence factor (dB) for interfering emissions in the reference bandwidth; it is positive when the interfering emissions would cause more degradation than thermal noise (see § 2.4 of this Annex).

In certain cases, an administration may have reason to believe that, for its receiving earth station, a departure from the values associated with the earth station, as listed in Table 8, may be justified. Attention is drawn to the fact that for specific systems the bandwidths B or, for example in the case of demand assignment systems, the percentages of the time p and p_0 may have to be changed from the values given in Table 8.

2.1 Calculation of the noise temperature of the receiving system

The noise temperature (K) of the receiving system, referred to the output terminals of the receiving antenna, may be determined (unless specifically given in Table 7) from:

$$T_e = T_a + (\ell_{t1} - 1)\,290 + \ell_{t1}\,T_r \quad \text{K} \tag{128}$$

where:

T_a: noise temperature (K) contributed by the receiving antenna

ℓ_{t1}: numerical loss in the transmission line (e.g. a waveguide) between the antenna terminal and the receiver front end

T_r: noise temperature (K) of the receiver front end, including all successive stages at the front end input.

For radio-relay receivers and where the waveguide loss of a receiving earth station is not known, a value of $\ell_{t1} = 1.0$ is used.

In case of determination of the coordination contours between two earth stations operating in the opposite direction of transmission, the following earth station receiving system noise temperatures should be used if the value is not provided in Table 9. This assumption is necessary because the receiving earth station takes the place of a receiving terrestrial station in the calculations.

TABLE 6

Frequency range (GHz)	T_e **(K)**
$f < 10$	75
$10 < f < 17$	150
$f > 17$	300

2.2 Determination of the factor N_L

The factor N_L is the noise contribution to the link. In the case of a satellite transponder, it includes the uplink noise, intermodulation, etc. In the absence of table entries, it is assumed:

N_L = 1 dB for fixed-satellite links

= 0 dB for terrestrial links

2.3 Determination of the factor M_S

The factor M_s is the factor by which the link noise under clear-sky conditions would have to be raised in order to equal the permissible interference power.

2.4 Determination of the factor W

The factor W (dB) is the level of the radio-frequency thermal noise power relative to the received power of an interfering emission which, in the place of the former and contained in the same (reference) bandwidth, would produce the same interference (e.g. an increase in the voice or video channel noise power, or in the bit error ratio). The factor W generally depends on the characteristics of both the wanted and the interfering signals.

When the wanted signal is digital, W is usually equal to or less than 0 dB, regardless of the characteristics of the interfering signal.

3 Horizon antenna gain for a receiving earth station with respect to a transmitting earth station

For the determination of the coordination area of a transmitting earth station with respect to a receiving earth station in a bidirectionally allocated band, it is necessary to calculate the horizon antenna gain of the unknown earth station. In cases where the unknown receiving earth stations operate with geostationary satellites, Table 9 provides the necessary receiving earth station parameters for the calculation procedure, which is described in § 2.1 of Annex 5.

In the case where the unknown receiving earth station operates with non-geostationary satellites, the horizon antenna gain to be used for all azimuths is provided in Table 9. The tabulated values were determined by using the method described in § 2.2 of the main body of this Appendix, which uses the maximum and minimum values of horizon antenna gain. For this purpose the maximum horizon antenna gain is the gain of the antenna for an off-axis angle equal to the minimum operating elevation angle. The minimum horizon antenna gain is the gain at large off-axis angles, usually more than 36° or 48°.

In determining the TIG horizon antenna gain entries in Table 9, the difference between the maximum and minimum horizon antenna gain did not exceed 30 dB. Consequently, the TIG horizon antenna gain was taken as the lesser of the maximum horizon antenna gain or 20 dB more than the minimum horizon antenna gain. For the purpose of determining the TIG horizon antenna gain, the reference antenna pattern of § 3 of Annex 3 was used, except in cases noted in the Tables where a different pattern was deemed to be more appropriate.

TABLE 7a (Rev.WRC-12)

Parameters required for the determination of coordination distance for a transmitting earth station

Transmitting space radiocommunication service designation		**Mobile-satellite, space operation**	**Earth exploration-satellite, meteorological satellite**		**Space operation**	**Space research, space operation**	**Mobile-satellite**	**Space operation**		**Mobile-satellite, radio-determination-satellite**	**Mobile-satellite**		**Space operation, space research**		**Mobile-satellite**		**Space research, space operation, Earth exploration-satellite**
Frequency bands (MHz)		148.0-149.9	401-403		433.75-434.25	449.75-450.25	806-840	1 427-1 429		1 610-1 626.5	1 668.4-1 675		1 750-1 850		1 980-2 025		2 025-2 110 2 110-2 120 (Deep space)
Receiving terrestrial service designations		Fixed, mobile	Meteorological aids		Amateur, radiolocation fixed, mobile	Fixed, mobile, radio-location	Fixed, mobile broadcasting, aeronautical radionavigation	Fixed, mobile		Aeronautical radionavigation	Fixed, mobile		Fixed, mobile		Fixed, mobile		Fixed, mobile
Method to be used		§ 2.1, § 2.2	§ 2.1, § 2.2		§ 2.1, § 2.2	§ 2.1, § 2.2	§ 1.4.6	§ 2.1, § 2.2		§ 1.4.6	§ 1.4.6		§ 2.1, § 2.2		§ 1.4.6		§ 2.1, § 2.2
Modulation at terrestrial station [1]		A	A	N		A and N	A and N	A	N		A	N	A	N	A	N	A
Terrestrial station interference parameters and criteria	p_0 (%)	1.0				0.01	0.01	0.01	0.01		0.01	0.01	0.01	0.01	0.01		0.01
	N	1				2	2	2	2		2	2	2	2	2		2
	p (%)	1.0				0.005	0.005	0.005	0.005		0.005	0.005	0.005	0.005	0.005		0.005
	N_L (dB)	–				0	0	0	0		0	0	0	0	0		0
	M_S (dB)	–				20	20	33	33		33	33	33	33	26 [2]		26 [2]
	W (dB)	–				0	0	0	0		0	0	0	0	0		0
Terrestrial station parameters	G_x (dBi) [3]	8				16	16	33	33		35	35	35	35	49 [2]		49 [2]
	T_e (K)	–				750	750	750	750		750	750	750	750	500 [2]		500 [2]
Reference bandwidth	B (Hz)	4×10^3				12.5×10^3	12.5×10^3	4×10^3	10^6		4×10^3	10^6	4×10^3	10^6	4×10^3		4×10^3
Permissible interference power	$P_r(p)$ (dBW) in B	−153				−139	−139	−131	−107		−131	−107	−131	−107	−140		−140

[1] A: analogue modulation; N: digital modulation.

[2] The parameters for the terrestrial station associated with transhorizon systems have been used. Line-of-sight radio-relay parameters associated with the frequency band 1 668.4-1 675 MHz may also be used to determine a supplementary contour. (WRC-03)

[3] Feeder losses are not included.

TABLE 7b (Rev.WRC-23)

Parameters required for the determination of coordination distance for a transmitting earth station

Transmitting space radiocommunication service designation		**Fixed-satellite, mobile-satellite**	**Aeronautical mobile-satellite (R) service**	**Aeronautical mobile-satellite (R) service**	**Fixed-satellite**	**Fixed-satellite**	**Fixed-satellite**	**Fixed-satellite**		**Earth exploration-satellite, space operation, space research**		**Fixed-satellite, mobile-satellite, meteorological-satellite**		**Fixed-satellite**		**Fixed-satellite**		**Fixed-satellite**	**Space research**		**Fixed-satellite** [3]	**Fixed-satellite**	**Fixed-satellite** [3]
Frequency bands (GHz)		2.655-2.690	5.030-5.091	5.030-5.091	5.091-5.150	5.091-5.150	5.725-5.850	5.725-7.075		7.100-7.250 [5]		7.900-8.400		10.7-11.7		12.5-14.8		13.75-14.3	14.8-15.35		15.43-15.65	17.7-18.4	19.3-19.7
Receiving terrestrial service designations		Fixed, mobile	Aeronautical radio-navigation	Aeronautical mobile (R)	Aeronautical radio-navigation	Aeronautical mobile (R)	Radiolocation	Fixed, mobile		Fixed, mobile		Fixed, mobile		Fixed, mobile		Fixed, mobile		Radiolocation radionavigation (land only)	Fixed, mobile		Aeronautical radionavigation	Fixed, mobile	Fixed, mobile
Method to be used		§ 2.1	§ 2.1, § 2.2	§ 2.1, § 2.2			§ 2.1	§ 2.1		§ 2.1, § 2.2		§ 2.1		§ 2.1		§ 2.1, § 2.2		§ 2.1	§ 2.1, § 2.2			§ 2.1, § 2.2	§ 2.2
Modulation at terrestrial station [1]		A						A	N	A	N	A	N	A	N	A	N	–	A	N		N	N
Terrestrial station interference parameters and criteria	p_0 (%)	0.01						0.01	0.005	0.01	0.005	0.01	0.005	0.01	0.005	0.01	0.005	0.01	0,01	0,005		0.005	0.005
	n	2						2	2	2	2	2	2	2	2	2	2	1	2	2		2	2
	p (%)	0.005						0.005	0.0025	0.005	0.0025	0.005	0.0025	0.005	0.0025	0.005	0.0025	0.01	0,005	0,0025		0.0025	0.0025
	N_L (dB)	0						0	0	0	0	0	0	0	0	0	0	0	0	0		0	0
	M_s (dB)	26 [2]						33	37	33	37	33	37	33	40	33	40	1	33	40		25	25
	W (dB)	0						0	0	0	0	0	0	0	0	0	0	0	0	0		0	0
Terrestrial station parameters	G_x (dBi) [4]	49 [2]	6	10	6	6		46	46	46	46	46	46	50	50	52	52	36	52	52		48	48
	T_e (K)	500 [2]						750	750	750	750	750	750	1 500	1 100	1 500	1 100	2 636	1 500	1 100		1 100	1 100
Reference bandwidth	B (Hz)	4×10^3	150×10^3	37.5×10^3	150×10^3	10^6		4×10^3	10^6	4×10^3	10^6	4×10^3	10^6	4×10^3	10^6	4×10^3	10^6	10^7	4×10^3	10^6		10^6	10^6
Permissible interference power	$P_r(p)$ (dBW) in B	−140	−160	−157	−160	−143		−131	−103	−131	−103	−131	−103	−128	−98	−128	−98	−131	−128	−98		−113	−113

[1] A: analogue modulation; N: digital modulation.

[2] The parameters for the terrestrial station associated with transhorizon systems have been used. Line-of-sight radio-relay parameters associated with the frequency band 5 725-7 075 MHz may also be used to determine a supplementary contour with the exception that $G_X = 37$ dBi.

[3] Feeder links of non-geostationary satellite systems in the mobile-satellite service.

[4] Feeder losses are not included.

[5] Actual frequency bands are 7 190-7 250 MHz for the Earth exploration-satellite service, 7 100-7 155 MHz and 7 190-7 235 MHz for the space operation service and 7 145-7 235 MHz for the space research service.

TABLE 7c (Rev.WRC-19)

Parameters required for the determination of coordination distance for a transmitting earth station

Transmitting space radiocommunication service designation		**Fixed-satellite**	**Fixed-satellite**	**Fixed-satellite** [2]	**Fixed-satellite**	**Space research**	**Earth exploration-satellite, space research**	**Fixed-satellite, mobile-satellite, radionavigation-satellite**	**Fixed-satellite** [2]	**Fixed-satellite**
Frequency bands (GHz)		24.65-25.25 27.0-29.5	24.65-25.25 27-27.5	28.6-29.1	27.5-28.6 29.1-29.5[3]	34.2-34.7	40.0-40.5	42.5-47 47.2-50.2 50.4-51.4	47.2-50.2	51.4-52.4
Receiving terrestrial service designations		Fixed (except HAPS), mobile	Fixed (HAPS ground station)	Fixed, mobile	Fixed, mobile	Fixed, mobile, radiolocation	Fixed, mobile	Fixed, mobile, radionavigation	Fixed, mobile	Fixed, mobile
Method to be used		§ 2.1	§ 2.1	§ 2.2	§ 2.2		§ 2.1, § 2.2	§ 2.1, § 2.2	§ 2.2	§ 2.1
Modulation at terrestrial station [1]		N	N	N	N		N	N	N	N
Terrestrial station interference parameters and criteria	p_0 (%)	0.005	0.01	0.005	0.005		0.005	0.005	0.001	0.005
	n	1	1	2	1		1	1	1	1
	p (%)	0.005	0.005	0.0025	0.005		0.005	0.005	0.001	0.005
	N_L (dB)	0	0	0	0		0	0	0	0
	M_s (dB)	25	10	25	25		25	25	25	25
	W (dB)	0	0	0	0		0	0	0	0
Terrestrial station parameters	G_x (dBi) [4]	50	0[5]	50	50		42	42	46	42
	T_e (K)	2 000	350	2 000	2 000		2 600	2 600	2 000	2 600
Reference bandwidth	B (Hz)	10^6	10^6	10^6	10^6		10^6	10^6	10^6	10^6
Permissible interference power	$P_r(p)$ (dBW) in B	−111	−134	−111	−111		−110	−110	−111	−110

[1] A: analogue modulation; N: digital modulation.

[2] Non-geostationary satellites in the fixed-satellite service.

[3] Feeder links to non-geostationary-satellite systems in the mobile-satellite service.

[4] Feeder losses are not included.

[5] Maximum HAPS ground station antenna gain toward the horizon.

TABLE 8a (Rev.WRC-23)

Parameters required for the determination of coordination distance for a receiving earth station

Receiving space radiocommunication service designation			**Space operation, space research**	**Meteoro-logical-satellite, mobile-satellite**	**Space research**	**Space research, space operation**	**Space operation**	**Mobile-satellite**	**Meteoro-logical-satellite**	**Mobile-satellite**	**Space research**	**Space operation**	**Meteoro-logical-satellite**	**Mobile-satellite**	**Broadcasting-satellite (DAB)**	**Mobile-satellite, land-mobile satellite, maritime mobile-satellite**
Frequency bands (MHz)			137-138	137-138	143.6-143.65	174-184	163-167 272-273 [5]	335.4-399.9	400.15-401	400.15-401	400.15-401	401-402	460-470	856-890	1 452-1 492	1 518-1 530 1 555-1 559 2 160-2 200 [1]
Transmitting terrestrial service designations			Fixed, mobile	Fixed, mobile	Fixed, mobile, radio-location	Fixed, mobile, broad-casting	Fixed, mobile	Fixed, mobile	Meteoro-logical aids	Meteoro-logical aids	Meteoro-logical aids	Meteoro-logical aids, fixed, mobile	Fixed, mobile	Fixed, mobile, broad casting	Fixed, mobile, broadcasting	Fixed, mobile
Method to be used			§ 2.1	§ 2.1	§ 2.1	§ 2.1	§ 2.1	§ 1.4.6	§ 1.4.6	§ 1.4.6	–	§ 2.1	§ 2.1	§ 1.4.6	§ 1.4.5	§ 1.4.6
Modulation at earth station [2]			N		N		N				N	N			N	N
Earth station interference parameters and criteria	p_0 (%)		0.1		0.1		1.0		0.012		0.1	0.1	0.012			10
	n		2		2		1		1		2	2	1			1
	p (%)		0.05		0.05		1.0		0.012		0.05	0.05	0.012			10
	N_L (dB)		0		0		0		0		0	0				0
	M_s (dB)		1		1		1		4.3		1	1				1
	W (dB)		0		0		0		0		0	0				0
Terrestrial station parameters	E (dBW) in B [3]	A	–		–		15				–	–	5		38	37 [4]
		N	–		–		15				–	–	5		38	37
	P_t (dBW) in B	A	–		–		–1				–	–	–11		3	0
		N	–		–		–1				–	–	–11		3	0
	G_x (dBi)		–		–		16				–	–	16		35	37
Reference bandwidth	B (Hz)		1		1		10^3		177.5×10^3		1	1	85		25×10^3	4×10^3
Permissible interference power	$P_r(p)$ (dBW) in B		−199		−199		−173		−148		−208	−208	−178			−176

[1] In the band 2 160-2 200 MHz, the terrestrial station parameters of line-of-sight radio-relay systems have been used. If an administration believes that, in this band transhorizon systems need to be considered, the parameters associated with the frequency band 2 500-2 690 MHz may be used to determine the coordination area.

[2] A: analogue modulation; N: digital modulation.

[3] E is defined as the equivalent isotropically radiated power of the interfering terrestrial station in the reference bandwidth.

[4] This value is reduced from the nominal value of 50 dBW for the purposes of determination of coordination area, recognizing the low probability of high power emissions falling fully within the relatively narrow bandwidth of the earth station.

[5] The fixed-service parameters provided in the column for 163-167 MHz and 272-273 MHz are only applicable to the band 163-167 MHz.

TABLE 8b (Rev.WRC-12)

Parameters required for the determination of coordination distance for a receiving earth station

Receiving space radiocommunication service designation			Space operation (GSO and non-GSO)	Meteoro-logical-satellite (non-GSO)	Meteoro-logical-satellite (GSO)	Space research near-Earth (non-GSO and GSO)		Space research deep space (non-GSO)	Space operation (non-GSO and GSO)	Earth exploration-satellite (GSO)	Broadcasting-satellite	Mobile-satellite, radio-determination-satellite	Fixed-satellite, broadcasting satellite		Fixed-satellite	
						Unmanned	Manned									
Frequency bands (GHz)			1.525-1.535	1.670-1.710	1.670-1.710	1.700-1.710 2.200-2.290		2.290-2.300	2.200-2.290	2.200-2.290	2.310-2.360	2.4835-2.500 [6]	2.500-2.690		3.400-4.200	
Transmitting terrestrial service designations			Fixed	Fixed, mobile, meteoro-logical aids	Fixed, mobile, meteoro-logical aids	Fixed, mobile		Fixed, mobile	Fixed, mobile	Fixed, mobile	Fixed, mobile, radiolocation	Fixed, mobile, radiolocation	Fixed, mobile radiolocation		Fixed, mobile	
Method to be used			§ 2.1, § 2.2	§ 2.2 and [1]	§ 2.1 and [1]	§ 2.1, § 2.2		§ 2.2	§ 2.1, § 2.2	§ 2.1	§ 1.4.5	§ 1.4.6	§ 1.4.5 and § 2.1		§ 2.1	
Modulation at earth station [2]			N	N	N	N		N	N	N		N	A	N	A	N
Earth station interference parameters and criteria	p_0 (%)		1.0	0.006	0.011	0.1	0.001	0.001	1.0	1.0		10	0.03	0.003	0.03	0.005
	n		1	3	2	2	1	1	2	2		1	3	3	3	3
	p (%)		1.0	0.002	0.0055	0.05	0.001	0.001	0.5	0.5		10	0.01	0.001	0.01	0.0017
	N_L (dB)		0	0	0	0		0	0			0	1	1	1	1
	M_s (dB)		1	2.8	0.9	1		0.5	1			1	7	2	7	2
	W (dB)		0	0	0	0		0	0			0	4	0	4	0
Terrestrial station parameters	E (dBW) in B [3]	A	50	92 [4]	92 [4]	−27 [4,5]		−27 [5]	72	72 [4]		37	72 [4]	72 [4]	55	55
		N	37	–	–	−27		−27	76	76		37	76	76	42	42
	P_t (dBW) in B	A	13	40 [4]	40 [4]	−71 [4,5]		−71 [5]	28	28 [4]		0	28 [4]	28 [4]	13	13
		N	0	–	–	−71		−71	32	32		0	32	32	0	0
	G_x (dBi)		37	52	52	44		44	44	44		37	44	44	42	42
Reference bandwidth	B (Hz)		10^3	10^6	4×10^3	1		1	10^6	10^6		4×10^3	10^6	10^6	10^6	10^6
Permissible interference power	$P_r(p)$ (dBW) in B		−184	−142	−177	−216		−222	−154	−154		−176				

[1] See Table 10.

[2] A: analogue modulation; N: digital modulation.

[3] E is defined as the equivalent isotropically radiated power of the interfering terrestrial station in the reference bandwidth.

[4] In this band, the parameters for the terrestrial stations associated with transhorizon systems have been used. If an administration believes that transhorizon systems do not need to be considered, the line-of-sight radio-relay parameters associated with the frequency band 3.4-4.2 GHz may be used to determine the coordination area, with the exception that E = 50 dBW for analogue terrestrial stations; and G_x = 37 dBi. However, for the space research service only, noting footnote 5 when transhorizon systems are not considered, E = 20 dBW and P_t = −17 dBW for analogue terrestrial stations, E = −23 dBW and P_t = −60 dBW for digital terrestrial stations; and G_x = 37 dBi.

[5] These values are estimated for 1 Hz bandwidth and are 30 dB below the total power assumed for emission.

[6] In the band 2.4835-2.5 GHz the terrestrial station parameters of line-of-sight radio-relay systems have been used. If an administration believes that, in this band, transhorizon systems need to be considered, the parameters associated with the frequency band 2 500-2 690 MHz may be used to determine the coordination area.

TABLE 8c (Rev.WRC-23)

Parameters required for the determination of coordination distance for a receiving earth station

Receiving space radiocommunication service designation			**Fixed-satellite**		**Fixed-satellite, radio-determination satellite**	**Fixed-satellite**	**Fixed-satellite**		**Meteoro-logical-satellite** [7, 8]	**Meteoro-logical-satellite** [9]	**Earth exploration-satellite** [7]	**Earth exploration-satellite** [9]	**Space research** [10]		**Fixed-satellite**		**Broadcasting-satellite**		**Space research**	**Broadcasting-satellite**	**Fixed-satellite** [7]
													Deep space								
Frequency bands (GHz)			4.500-4.800		5.150-5.216	6.700-7.075	7.250-7.750		7.450-7.550	7.750-7.900	8.025-8.400	8.025-8.400	8.400-8.450	8.450-8.500	10.7-12.75 13.4-13.65 [7]		12.5-12.75 [12]		14.8-15.35	17.7-17.8	17.7-18.8 19.3-19.7
Transmitting terrestrial service designations			Fixed, mobile		Aeronautical radionavigation	Fixed, mobile	Fixed, mobile		Fixed, mobile	Fixed, mobile	Fixed, mobile	Fixed, mobile	Fixed, mobile		Fixed, mobile		Fixed, mobile		Fixed, mobile	Fixed	Fixed, mobile
Method to be used			§ 2.1		§ 2.1	§ 2.2	§ 2.1		§ 2.1, § 2.2	§ 2.2	§ 2.1	§ 2.2	§ 2.2		§ 2.1, § 2.2		§ 1.4.5		§ 2.1, § 2.2	§ 1.4.5	§ 2.1
Modulation at earth station [1]			A	N		N	A	N	N	N	N	N	N	N	A	N	A	N	N		N
Earth station interference parameters and criteria	p_0 (%)		0.03	0.005		0.005	0.03	0.005	0.002	0.001	0.083	0.011	0.001	0.1	0.03	0.003	0.03	0.003	0.1		0.003
	n		3	3		3	3	3	2	2	2	2	1	2	2	2	1	1	2		2
	p (%)		0.01	0.0017		0.0017	0.01	0.0017	0.001	0.0005	0.0415	0.0055	0.001	0.05	0.015	0.0015	0.03	0.003	0.05		0.0015
	N_L (dB)		1	1		1	1	1	–	–	1	0	0	0	1	1	1	1	0		1
	M_s (dB)		7	2		2	7	2	–	–	2	4.7	0.5	1	7	4	7	4	1		6
	W (dB)		4	0		0	4	0	–	–	0	0	0	0	4	0	4	0	0		0
Terrestrial station parameters	E (dBW) in B [2]	A	92 [3]	92 [3]		55	55	55	55	55	55	55	25 [5]	25 [5]	40	40	55	55	32		35
		N	42 [4]	42 [4]		42	42	42	42	42	42	42	−18	−18	43	43	42	42	40	40	40
	P_t (dBW) in B	A	40 [3]	40 [3]		13	13	13	13	13	13	13	−17 [5]	−17 [5]	−5	−5	10	10	−5		−10
		N	0	0		0	0	0	0	0	0	0	−60	−60	−2	−2	−3	−3	−5	−7	−5
	G_x (dBi)		52 [3, 4]	52 [3, 4]		42	42	42	42	42	42	42	42	42	45	45	45	45	35	47	45
Reference bandwidth [6]	B (Hz)		10^6	10^6		10^6	10^6	10^6	10^7	10^7	10^6	10^6	1	1	10^6	10^6	27×10^6	27×10^6	10^6		10^6
Permissible interference power	$P_r(p)$ (dBW) in B					−151.2			−125	−125	−154 [11]	−142	−220	−216			−131	−131	−156		

Notes to Table 8c:

1 A: analogue modulation; N: digital modulation.

2 E is defined as the equivalent isotropically radiated power of the interfering terrestrial station in the reference bandwidth.

3 In this band, the parameters for the terrestrial stations associated with transhorizon systems have been used. If an administration believes that transhorizon systems do not need to be considered, the line-of-sight radio-relay parameters associated with the frequency band 3.4-4.2 GHz may be used to determine the coordination area.

4 Digital systems assumed to be non-transhorizon. Therefore $G_x = 42.0$ dBi. For digital transhorizon systems, parameters for analogue transhorizon systems above have been used.

5 These values are estimated for 1 Hz bandwidth and are 30 dB below the total power assumed for emission.

6 In certain systems in the fixed-satellite service it may be desirable to choose a greater reference bandwidth B. However, a greater bandwidth will result in smaller coordination distances and a later decision to reduce the reference bandwidth may require recoordination of the earth station.

7 Geostationary-satellite systems.

8 Non-geostationary satellites in the meteorological-satellite service notified in accordance with No. **5.461A** may use the same coordination parameters.

9 Non-geostationary satellite systems.

10 Space research earth stations in the frequency band 8.4-8.5 GHz operate with non-geostationary satellites.

11

For large earth stations:	$P_x(p) = (G - 180)$	dBW	
For small earth stations:	$P_x(20\%) = 2\,(G - 26) - 140$	dBW	for $26 < G \leq 29$ dBi
	$P_x(20\%) = G - 163$	dBW	for $G > 29$ dBi
	$P_x(p)\% = G - 163$	dBW	for $G \leq 26$ dBi

12 Applies to the broadcasting-satellite service in unplanned bands in Region 3.

TABLE 8d (Rev.WRC-19)

Parameters required for the determination of coordination distance for a receiving earth station

Receiving space radiocommunication service designation			Meteoro-logical-satellite	Fixed-satellite	Fixed-satellite [3]	Broad-casting-satellite	Earth exploration-satellite [4]	Earth exploration-satellite [5]	Space research (deep space)	Space research		Fixed-satellite [6]	Fixed-satellite [5]	Mobile-satellite	Broadcasting-satellite, fixed-satellite	Mobile-satellite	Radio-navigation-satellite
										Unmanned	Manned						
Frequency bands (GHz)			18.0-18.4	17.8-18.6 [4] 18.8-19.3	19.3-19.7	21.4-22.0	25.5-27.0	25.5-27.0	31.8-32.3	37.0-38.0		37.5-40.5	37.5-40.5	39.5-40.5	40.5-42.5	43.5-47.0	43.5-47.0
Transmitting terrestrial service designations			Fixed, mobile	Fixed, mobile	Fixed, mobile	Fixed, mobile	Fixed, mobile	Fixed, mobile	Fixed, radio-navigation	Fixed, mobile		Fixed, mobile	Fixed, mobile	Fixed, mobile	Broadcasting, fixed	Mobile	Mobile
Method to be used			§ 2.1	§ 2.1, § 2.2	§ 2.2	§ 1.4.5	§ 2.2	§ 2.1	§ 2.1, § 2.2	§ 2.1, § 2.2		§ 2.2	§ 2.1	§ 1.4.6	§ 1.4.5, § 2.1	§ 1.4.6	–
Modulation at earth station [1]			N	N	N		N	N	N	N		N	N	N	–	N	
Earth station interference parameters and criteria	p_0 (%)		0.05	0.003	0.01		0.25	0.25	0.001	0.1	0.001	0.02	0.003				
	n		2	2	1		2	2	1	1	1		2				
	p (%)		0.025	0.0015	0.01		0.125	0.125	0.001	0.1	0.001		0.0015				
	N_L (dB)		0	0	0		0	0	0	0		1	1				
	M_S (dB)		18.8	5	5		11.4	14	1	1		6.8	6				
	W (dB)		0	0	0		0	0	0	0		0	0				
Terrestrial station parameters	E (dBW) in B [2]	A		–	–		–	–	–	–		–	–	–	–		
		N	40	40	40	40	42	42	−28	−28		35	35	35	44	40	40
	P_t (dBW) in B	A		–	–		–	–	–	–		–	–	–	–		
		N	−7	−7	−7	−7	−3	−3	−81	−73		−10	−10	−10	−1	−7	−7
	G_x (dBi)		47	47	47	47	45	45	53	45		45	45	45	45	47	47
Reference bandwidth [7]	B (Hz)		10^7	10^6	10^6		10^7	10^7	1	1		10^6	10^6	10^6	10^6		
Permissible interference power	$P_r(p)$ (dBW) in B		−115	−140	−137		−120	−116	−216	−217		−140					

[1] A: analogue modulation; N: digital modulation.

[2] E is defined as the equivalent isotropically radiated power of the interfering terrestrial station in the reference bandwidth.

[3] Non-geostationary mobile-satellite service feeder links.

[4] Non-geostationary-satellite systems.

[5] Geostationary-satellite systems.

[6] Non-geostationary fixed-satellite service systems.

[7] In certain systems in the fixed-satellite service it may be desirable to choose a greater reference bandwidth B. However, a greater bandwidth will result in smaller coordination distances, and a later decision to reduce the reference bandwidth may require recoordination of the earth station.

TABLE 9a (Rev.WRC-19)

Parameters required for the determination of coordination distance for a transmitting earth station in bands shared bidirectionally with receiving earth stations

Space service designation in which the transmitting earth station operates		**Mobile-satellite**	**Earth exploration-satellite, meteorological-satellite**	**Mobile-satellite**		**Fixed-satellite, mobile-satellite**	**Aeronautical mobile-satellite (R) service**		**Fixed-satellite [3]**		**Fixed-satellite**	**Fixed-satellite, meteorological-satellite**	**Fixed-satellite**
Frequency bands (GHz)		0.272-0.273	0.401-0.402	1.670-1.675		2.655-2.690	5.030-5.091		5.150-5.216		6.700-7.075	8.025-8.400	8.025-8.400
Space service designation in which the *receiving* earth station operates		Space operation	Space operation	Meteorological-satellite		Fixed-satellite, broadcasting-satellite	Aeronautical mobile-satellite (R) service		Fixed-satellite	Radiodetermination-satellite	Fixed-satellite	Earth exploration-satellite	Earth exploration-satellite
Orbit [6]		Non-GSO	Non-GSO	Non-GSO	GSO		Non-GSO	GSO	Non-GSO		Non-GSO	Non-GSO	GSO
Modulation at *receiving* earth station [1]		N	N	N	N						N	N	N
Receiving earth station interference parameters and criteria	p_0 (%)	1.0	0.1	0.006	0.011						0.005	0.011	0.083
	n	1	2	3	2						3	2	2
	p (%)	1.0	0.05	0.002	0.0055						0.0017	0.0055	0.0415
	N_L (dB)	0	0	0	0						1	0	1
	M_s (dB)	1	1	2.8	0.9	2			2	2	2	4.7	2
	W (dB)	0	0	0	0						0	0	0
Receiving earth station parameters	G_m (dBi) [2]	20	20	30	45		45	45	48.5		50.7		
	G_r (dBi) [4]	19	19	19 [9]	8		8	8	10		10	10	8
	ε_{min} [5]	10°	10°	5°	3°	3°	10°	10°	3°	3°	3°	5°	3°
	T_e (K) [7]	500	500	370	118	75	340	340	75	75	75		
Reference bandwidth	B (Hz)	10^3	1	10^6	4×10^3		37.5×10^3	37.5×10^3			10^6	10^6	10^6
Permissible interference power	$P_r(p)$ (dBW) in B	−177	−208	−145	−178		−163.5	−163.5			−151	−142	−154

Notes to Table 9a:

1 A: analogue modulation; N: digital modulation.

2 On-axis gain of the receive earth station antenna.

3 Feeder links of non-geostationary-satellite systems in the mobile-satellite service.

4 Horizon antenna gain for the receive earth station (refer to § 3 of the main body of this Appendix and to § 3 of this Annex).

5 Minimum elevation angle of operation in degrees (non-geostationary or geostationary).

6 Orbit of the space service in which the receiving earth station operates (non-geostationary or geostationary).

7 The thermal noise temperature of the receiving system at the terminal of the receiving antenna (under clear-sky conditions). Refer to § 2.1 of this Annex for missing values.

8 Horizon antenna gain is calculated using the procedure of Annex 5. Where no value of G_m is specified, a value of 42 dBi is to be used.

9 Non-geostationary horizon antenna gain, $G_e = G_{min} + 20$ dB (see § 2.2), with $G_{min} = 10 - 10 \log (D/\lambda)$, $D/\lambda = 13$ (refer to Annex 3 for definition of symbols).

TABLE 9b (Rev.WRC-19)

Parameters required for the determination of coordination distance for a transmitting earth station in bands shared bidirectionally with receiving earth stations

Space service designation in which the transmitting earth station operates		**Fixed-satellite**			**Fixed-satellite**			**Fixed-satellite**	**Fixed-satellite**	**Fixed-satellite [3]**	**Fixed-satellite [3]**	**Earth exploration-satellite, space research**	
Frequency bands (GHz)		10.7-11.7			12.5-12.75			17.3-17.8	17.7-18.4	19.3-19.6	19.3-19.6	40.0-40.5	
Space service designation in which the *receiving* earth station operates		Fixed-satellite			Fixed-satellite			Broadcasting-satellite	Fixed-satellite, meteorological-satellite	Fixed-satellite [3]	Fixed-satellite [4]	Fixed-satellite, mobile-satellite	
Orbit [7]		GSO		Non-GSO	GSO		Non-GSO		GSO	Non-GSO	GSO	GSO	Non-GSO
Modulation at *receiving* earth station [1]		A	N	N	A	N			N	N			
Receiving earth station interference parameters and criteria	p_0 (%)	0.03	0.003		0.03	0.003			0.003	0.01	0.003	0.003	
	N	2	2		2	2			2	1	2	2	
	p (%)	0.015	0.0015		0.015	0.0015			0.0015	0.01	0.0015	0.0015	
	N_L (dB)	1	1		1	1			1	0	1	1	
	M_s (dB)	7	4		7	4			6	5	6	6	
	W (dB)	4	0		4	0			0	0	0	0	
Receiving earth station parameters	G_m (dBi) [2]			51.9			31.2		58.6	53.2	49.5	50.8	54.4
	G_r (dBi) [5]	9	9	10	9	9	11 [11]		9	10	10	9	7 [12]
	ε_{min} [6]	5°	5°	6°	5°	5°	10°		5°	5°	10°	10°	10°
	T_e (K) [8]	150	150		150	150			300	300	300	300	
Reference bandwidth	B (Hz)	10^6	10^6		10^6	10^6			10^6	10^6			
Permissible interference power	$P_r(p)$ (dBW) in B	−144	−144	−144	−144	−144	−144		−138	−141			

Notes to Table 9b:

1 A: analogue modulation; N: digital modulation.

2 On-axis gain of the receive earth station antenna.

3 Feeder links of non-geostationary-satellite systems in the mobile-satellite service.

4 Geostationary-satellite systems.

5 Horizon antenna gain for the receive earth station (refer to § 3 of the main body of the Appendix and to § 3 of this Annex).

6 Minimum elevation angle of operation in degrees (non-GSO or GSO).

7 Orbit of the space service in which the receiving earth station operates (GSO or non-GSO).

8 The thermal noise temperature of the receiving system at the terminal of the receiving antenna (under clear-sky conditions). Refer to § 2.1 of this Annex for missing values.

9 Horizon antenna gain is calculated using the procedure of Annex 5. Where no value of G_m is specified, a value of 42 dBi is to be used.

10 Horizon antenna gain is calculated using the procedure of Annex 5, except that the following antenna pattern may be used in place of that given in § 3 of Annex 3: $G = 32 - 25 \log \varphi$ for $1° \leq \varphi < 48°$; and $G = -10$ for $48° \leq \varphi < 180°$ (refer to Annex 3 for definition of symbols).

11 Non-geostationary horizon antenna gain. $G_e = G_{max}$ (see § 2.2 of the main body of this Appendix) with the antenna pattern modelled by the equation $G = 36 - 25 \log (\varphi)$ (refer to Annex 3 for definition of symbols).

12 Non-geostationary horizon antenna gain. $G_e = G_{max}$ (see § 2.2 of the main body of this Appendix) with the antenna pattern modelled by the equation $G = 32 - 25 \log (\varphi)$ (refer to Annex 3 for definition of symbols).

TABLE 10 (Rev.WRC-19)

Predetermined coordination distances

Frequency sharing situation		Coordination distance (in sharing situations involving services allocated with equal rights) (km)
Type of earth station	**Type of terrestrial station**	
Ground-based in the bands below 1 GHz to which No. **9.11A** applies. Ground-based mobile in the bands within the range 1-3 GHz to which No. **9.11A** applies	Mobile (aircraft)	500
Aircraft (all bands)	Ground-based	500
Aircraft (all bands)	Mobile (aircraft)	1 000
Ground-based in the bands: 400.15-401 MHz 1 668.4-1 675 MHz	Station in the meteorological aids service (radiosonde)	580
Aircraft in the bands: 400.15-401 MHz 1 668.4-1 675 MHz	Station in the meteorological aids service (radiosonde)	1 080
Ground-based in the radiodetermination-satellite service (RDSS) in the bands: 1 610-1 626.5 MHz 2 483.5-2 500 MHz 2 500-2 516.5 MHz	Ground-based	100
Airborne earth station in the radiodetermination-satellite service (RDSS) in the bands: 1 610-1 626.5 MHz 2 483.5-2 500 MHz 2 500-2 516.5 MHz	Ground-based	400
Receiving earth stations in the meteorological-satellite service	Station in the meteorological aids service	The coordination distance is considered to be the visibility distance as a function of the earth station horizon elevation angle for a radiosonde at an altitude of 20 km above mean sea level, assuming 4/3 Earth radius (see Note 1)
Non-GSO MSS feeder-link earth stations (all bands)	Mobile (aircraft)	500
Non-GSO MSS feeder-link earth stations in the band 5 091-5 150 MHz	Station in the aeronautical radionavigation service	Note 2
Receiving earth stations in the space research service in the band: 2 200-2 290 MHz	Mobile (aircraft)	880
Ground-based in the bands in which the frequency sharing situation is not covered in the rows above	Mobile (aircraft)	500

Notes to Table 10:

NOTE 1 – The coordination distance, d (km), for fixed earth stations in the meteorological-satellite service vis-à-vis stations in the meteorological aids service assumes a radiosonde altitude of 20 km and is determined as a function of the physical horizon elevation angle ε_h (degrees) for each azimuth, as follows:

$d = 100$ for $\varepsilon_h \geq 11°$

$d = 582\left(\sqrt{1 + (0.254\ \varepsilon_h)^2} - 0.254\ \varepsilon_h\right)$ for $0° < \varepsilon_h < 11°$

$d = 582$ for $\varepsilon_h \leq 0°$

The minimum and maximum coordination distances are 100 km and 582 km, and correspond to physical horizon angles greater than 11° and less than 0°. (WRC-2000)

NOTE 2 – For the coordination distance in the frequency band 5 091-5 150 MHz vis-à-vis stations in the aeronautical radionavigation service, see No. **5.444A**. (WRC-15)